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DAY -1 PYTHON -DATA SCIENCE

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Data Types

Variables

In programming, **variables** are used for storing data values. Each variable has a name and holds a value. Think of it like a box: slap on a label, and it can store stuff. 📦

The variable name can consist of letters, numbers, and the _ underscore.

These are all valid variable names and values:

```
name = 'Erlich Bachman'
user_id = 16180339887
progress = 0.75
xp = 60
verified = True
```

The = equal sign means assignment:

- We're assigning the string value `'Erlich Bachman'` to the variable `name`.
- We're assigning the number value `16180339887` to the variable `user_id`.
- We're assigning the number value `0.75` to the variable `progress`.
- We're assigning the number value `60` to the variable `xp`.
- We're assigning the truth value `True` to the variable `verified`.

We can also change the value of a variable or print it out:

```
xp = 70
xp = 80

print(xp)     # Output: 80
```

Here, we are assigning the number value `70` to the variable `xp`. Then, we are reassigning the number value `80` to the same variable. And printing it out.

Data Types

Integer

An integer, or `int`, is a whole number. It has no decimal point and contains the number 0, positive and negative counting numbers. If we were counting the number of people on the bus or the number of jellybeans in a jar, we would use an integer.

```
year = 2023
age = 32
```

Float

A floating-point number, or a `float`, is a decimal number. It can be used to represent fractions or precise measurements. If we were measuring the length and width of the couch, calculating the test score percentage, or storing a baseball player's batting average, we would use a `float` instead of an `int`.

```
pi = 3.14159
meal_cost = 12.99
```

String

A string, or `str`, is used for storing text. Strings are wrapped in double quotes `" "` or single quotes `' '`.

```
message = "good nite"
username = '@snoopdogg'
```

When we did printing in the first chapter, we printed a string data type.

Boolean

A Boolean data type, or `bool`, stores a value that can only be either true or false. In Python, it's capitalized `True` or `False`. It's named after the British mathematician George Boole (1815-1864).

```
late_to_class = False
cranky = True
```